

Ananke: A Jupyter distribution integrating JupyterHub and Nbgrader into learning management systems

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Software

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Summary

The Jupyter ecosystem (Jupyter Development Team, 2024a) is widely used in STEM teaching. Major components include JupyterLab as the user-facing environment, JupyterHub for managing user access, and Nbgrader (Jupyter Development Team, 2024b) (originally described in (Jupyter et al., 2019)) for autograding assignments. The Ananke Jupyter distribution on the one hand bundles pre-configured tools from the Jupyter ecosystem for academic teaching and research. On the other hand, Ananke provides tight integration of JupyterHub and Nbgrader into learning management systems (LMS) like Moodle (Moodle Pty Ltd, 2024) and Canvas (Instructure Global Ltd, 2024) via the Learning Tools Interoperability (LTI) 1.3 standard. The aim of Ananke is to make installation and administration of server-based Jupyter deployments in academic environments as simple as possible. The focus is on small to medium-sized JupyterHubs running on one server, not on distributed large-scale ones.

Statement of need

The need for Ananke arose from the realization that the installation of each individual Jupyter component is easy, but the effort required for a practically usable overall system is very high. An even more difficult problem is that most universities run an LMS for managing students, grades and material. Thus, grading tools like Nbgrader should integrate seamlessly into the LMS. While JupyterHub already provides login through an LMS, Nbgrader requires that each course, instructor and student is manually enrolled to its database, which is far from suitable for everyday use. There's a standard for data transfer between LMS and external tools known as LTI (IMS Global Learning Consortium, 2019b, 2019c, 2019a, 2019d). So we developed the tool Kore, which handles all the data transfer. Due to its tight integration into JupyterHub and Nbgrader, it should not be considered a separate tool, but an integral part of the Ananke Jupyter distribution.

Due to its LTI abilities, with Ananke manual additions to Nbgrader's database are obsolete. Necessary information is copied on user login from the LMS to JupyterHub automatically. In addition, instructors may send all the grades generated by Nbgrader back to the LMS simply by clicking a button. Ananke thus greatly simplifies using Nbgrader in a multi-course and multi-instructor teaching environment.

Documentation and features

Looking for a way to document and automate all installation steps the authors built a Podman image containing all components and configuration files. Podman (Podman, 2024) is a Docker-like and Docker-compatible containerization system. Apart from ease of install,

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